TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable Chiral Emissive Materials Enabling Intrinsic Polarization Control for Next-Generation Displays			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Electrical and Electronic Components	Display Components and Materials	Optical Application Devices	Optical Materials
Concept of Proposed Technology	The proposed technology aims to fundamentally simplify conventional display architectures that rely on external polarizers or complex optical stacks by embedding polarization control directly into the emissive material itself. Specifically, this project seeks to develop a scalable and patternable chiral emissive material platform in which polarization-selective or circularly polarized emission is generated intrinsically at the emission layer. Conventional polarization-based displays—such as 3D displays, AR/VR/XR systems, and microdisplays incorporating polarization optics—typically employ external polarizers, waveplates, and multilayer optical films to control polarization states. These approaches inherently introduce significant optical losses, increased device thickness, elevated process complexity, and limited pixel-level design flexibility. In contrast, the proposed approach leverages chiral light–matter interactions in chiral metal and semiconductor nanomaterials (and their composite emissive layers) to generate circularly polarized or polarization-selective emission directly at the material level. Our research group has established proprietary capabilities in (i) scalable synthesis of chiral metal and semiconductor nanostructures [1-10], (ii) advanced spectroscopy-based polarization stage characterization [11,12] and (iii) processing strategies compatible with spatial patterning at the pixel and sub-pixel level [3]. Building upon this foundation, chiral nanomaterial can be integrated into OLEDs, quantum-dot LEDs, and perovskite LEDs, either as emissive layers or as color-conversion			

	and composite emissive layers. This strategy enables pixel-level polarization encoding and polarization-selective emission without reliance on external polarizers.
Rationale for Joint R&D Project	For chiral emissive materials to mature beyond laboratory-scale CPL/PL demonstrations and reach TRL 4 or higher (device- and module-level validation), several integration bottlenecks must be addressed that cannot be resolved through materials synthesis and patterning alone. In particular, display applications simultaneously require: (1) maintenance of polarization performance under electrically driven operation, (2) high-precision and uniform pixel patterning, (3) operational lifetime and reliability under thermal, electrical, and humidity stress, (4) compatibility with large-area manufacturing and existing fabrication lines, and (5) module-level evaluation incorporating optics, driving electronics, and packaging. These requirements are difficult to fulfill within a single academic laboratory environment. Accordingly, this international joint R&D project is structured to integrate materials, processes, devices, and modules into a continuous development chain that systematically addresses TRL bottlenecks. • The research institute (RPI) will lead the design and synthesis of chiral metal and semiconductor nanomaterials, optimization of chiroptical properties (PL/CPL), development of patterning-compatible inks and thin films, and architectural design of composite emissive layers that couple chirality with light emission, thereby establishing a robust scientific and materials foundation for intrinsically polarization-controlled emission. • The Korean SME partner will contribute expertise in electroluminescent device fabrication, electrical operation, and performance evaluation under display-relevant conditions. Through this collaboration, the project converts the fundamental limitations of external-polarizer-based display architectures—namely optical loss, thickness, and complexity—into a materials-based solution, enabling device-level validation while remaining fully aligned with the pre-commercial scope of TRL 4–5 research.
Key Words	1. Chiral emissive nanomaterial 2. Device Integration of Chiral Photonic Materials 3. Intrinsic Circularly Polarized Emission 4. Pixel-level polarization encoding

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2-3)	**End of the Project**	Level (4-5)
R&D Objectives	Next-generation displays require simultaneous improvements in optical efficiency, compactness, and functional integration. However, conventional polarization control schemes based on external polarizers and multilayer optical stacks introduce inherent optical losses and structural complexity. The overarching objective of this project is to develop a scalable and patternable chiral emissive material platform that enables intrinsic generation of circularly polarized or polarization-selective emission at the emissive layer, thereby eliminating or minimizing external polarizers and simplifying display architectures. Specifically, this project aims to achieve the following: • Material Level: High-efficiency CPL emission from chiral metal and semiconductor nanomaterials with uniform thin-film characteristics • Process Level: Compatibility with pixel-level patterning and large-area manufacturing through ink-based and thin-film processing			

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	• Device Level: Demonstration of circularly polarized electroluminescence (CP-EL) with meaningful polarization performance, operational stability, and lifetime without external polarizers • Module / Prototype Level: Establishment of reproducible material–device integration guidelines supporting future applied research
R&D Target Specifications	1. Chiroptical / Polarization Performance • CPL dissymmetry factor (g_lum): |g_lum| ≥ 0.05 (thin-film level; target ≥ 0.1) • Polarization contrast (PL): ≥ 1.5:1 (LCP:RCP or vice versa) • Thin-film uniformity: Spatial variation of g_lum ≤ ±10% over ≥ 2 × 2 cm² 2. Patternability / Manufacturability • Patterning resolution: ≤ 20 μm line/space • Process compatibility: Retention of CPL performance in ≥ two patterning processes (e.g., inkjet, spin coating, slot-die, photopatterning) • Film thickness window: 50–300 nm with ≤ 10% CPL variation 3. Device-Level Polarization Performance • Electroluminescent dissymmetry factor (g_EL): |g_EL| ≥ 0.10 • Operational stability: ≥ 80% polarization retention after 100 h operation • Reproducibility: consistent performance across multiple devices 4. Integration and Patterning Compatibility • Pixel-scale integration: demonstrated at ≥ 20 μm feature size • Processing compatibility: successful integration using at least two thin-film or solution-based processes • Thickness tolerance: stable performance across 50–300 nm emissive layers
R&D Plan	**Aim 1. Definition of Device-Relevant Integration Framework** (0–6 months) • Selection of representative electroluminescent device configurations • Definition of polarization and stability evaluation metrics • Establishment of joint experimental protocols Milestone: Device integration framework and evaluation criteria established **Aim 2. Optimization of Chiral Emissive Materials for Thin-Film Integration** (0–12 months) • Optimization of nanomaterial composition and surface chemistry • Thin-film fabrication and CPL characterization • Assessment of pixel-scale patterning compatibility Milestone: Thin films exhibiting |g_lum| ≥ 0.05 with spatial uniformity **Aim 3. Electroluminescent Device Demonstration** (7–24 months) • Integration into electroluminescent devices • Evaluation of polarization performance under electrical operation • Analysis of efficiency–polarization relationships Milestone: Device-level demonstration of |g_EL| ≥ 0.10 **Aim 4. Stability and Reproducibility Assessment** (19–30 months) • Continuous operation testing • Thermal and electrical stress evaluation • Identification of degradation mechanisms Milestone: ≥ 80% polarization retention after 100 h operation **Aim 5. TRL 4–5 Validation and Design Guideline Consolidation** (25–36 months) • Consolidation of material–device integration insights • Formulation of design and processing guidelines • Final TRL 4–5 validation summary

	Milestone: Device-level feasibility and applicability conclusively validated

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead the **materials-level design, synthesis, and fundamental optical investigation** of chiral emissive nanomaterials, providing the scientific foundation required for device-level validation. In particular, RPI will focus on understanding and optimizing the intrinsic relationship between material chirality, optical emission, and polarization behavior. • **Synthesis and optical optimization of chiral emissive nanomaterials** Development of chiral metal and semiconductor nanomaterials with controlled structural chirality, emission characteristics, and polarization response, including systematic optimization of composition, morphology, and surface chemistry. • **Thin-film integration and CPL characterization** Fabrication of uniform thin-film emissive layers and comprehensive characterization of circularly polarized luminescence (CPL), including dissymmetry factors, polarization contrast, and spatial uniformity relevant to device integration. • **Mechanistic analysis of polarization generation and degradation** Investigation of the physical and chemical mechanisms governing polarization generation, efficiency–polarization trade-offs, and degradation pathways under optical and electrical stress, providing design insights to support stable device operation.
Expected R&R of Korean SME Partner	The Korean SME partner will contribute expertise in **electroluminescent device integration and evaluation**, enabling assessment of chiral emissive materials under realistic operating conditions. The SME's role is essential for translating material-level advances into device-level feasibility demonstrations consistent with TRL 4–5 research. • **Electroluminescent device fabrication and electrical operation** Integration of chiral emissive thin films into representative electroluminescent device structures and execution of electrically driven operation to assess functional performance. • **Evaluation of polarization performance and stability** Measurement and analysis of polarization-controlled electroluminescence, operational stability, and device-to-device reproducibility under continuous electrical driving and thermal stress conditions. • **Feedback-driven refinement of integration strategies** Iterative feedback to the research institute regarding device integration challenges, enabling refinement of material formulations, film structures, and interface design for improved device compatibility.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to establish a **robust technical foundation** for advancing chiral emissive materials from material-level discovery toward **device-level applicability**, in alignment with TRL 4–5 research objectives. By systematically validating intrinsic polarization behavior under electrical operation, the project will reduce key technical uncertainties that currently limit the translation of chiral photonic materials into device-relevant contexts. **Expected Impact** • **Improved understanding of intrinsic polarization behavior under electrical excitation** Generation of fundamental insights into how chiral emissive materials produce and maintain

	polarization-controlled emission under electrically driven conditions, including the influence of device architecture and operating stress. • **Demonstration of feasibility for integrating chiral materials into display-relevant devices** Experimental validation that chiral emissive materials can be incorporated into electroluminescent devices while retaining polarization functionality, reproducibility, and operational stability. • **Clear guidance for future applied and translational research beyond TRL 5** Establishment of material–device integration principles and performance benchmarks that inform subsequent applied research and potential scale-up efforts without extending into commercialization.

6. References

(1) W.J. Choi, J.-Y. Kim M. Hwang, C.M. Lim, B.C Park, A Emre, S. H. Lee, N. A. Kotov, "Highly Elliptic Circular Dichroism of Copper Aspartate One-Dimensional Nanostructures across the Ultraviolet to Terahertz Ranges" ***Nano Letters*****,** 2025, 25, 7699-7706

(2) J. Lu, H. J. Jung, J.-Y. Kim, N. A. Kotov, "Bright, circularly polarized black-body radiation from twisted nanocarbon filaments" ***Science*****,** 2024, 386, pp1400-1404

(3) J.-Y. Kim, C. McGlothin, M. Cha, Z. J.Pfaffenberger, E. S. Turali-Emre, S. Lee, W. Choi, J. Biteen, N. A. Kotov, "Direct-write 3D printing of plasmonic nanohelicoids by circularly polarized light", ***Proceedings of the National Academy of Sciences*** 2024**,** 121, e2312082121

(4) P. Kumar, T. Vo, M. Cha, A. Visheratina, J.-Y. Kim, W. Xu, J. Schwartz, A. Simon, D. Katz, E. Marino, W. J. Choi, S. Chen, C. Murray, R. Hovden, S. Glotzer, N. A. Kotov*, "Photonically Active Bowtie Nanoassemblies with Chirality Continuum" ***Nature*** 2023, 615, 418-424

(5) S. Zhou, J. Li, J. Lu, H. Liu, J.-Y. Kim, A. Kim, L. Yao, Ch. Liu, C. Qian, Z. D. Hood, X. Lin, W. Chen, T. E. Gage, I. Arslan, A. Travesset, K. Sun, N. A. Kotov, Qian Chen, "Chiral Assemblies of Pinwheel Superlattices on Substrates", ***Nature*** 2022, 612, 259–265

(6) J. Guo, J.-Y. Kim, S. Yang, J. Xu, Y. C. Choi, A. Stein, C. B. Murray, N. A. Kotov, Cherie R Kagan, "Broadband Circular Polarizers via Coupling in 3D Plasmonic Meta-Atom Arrays", ***ACS Photonics*** 2021, 8, 1286-1292

(7) J. Ahn, S. Ma, J.-Y. Kim, W. Yang, N. A. Kotov, and J. Moon, "Chiral 2D Organic Inorganic Hybrid Perovskite with Circular Dichroism Tunable Over Wide Wavelength Range", ***J. Am. Chem. Soc.*** 2020, 142, 4206–4212

(8) J. Yan, W. Feng, J.-Y. Kim, Z. Mu, P. Kumar, X. Mao, X. Wu, N. A. Kotov, "Self-Assembly of Chiral Nanoparticles into Semiconductor Helices with Tunable Near-Infrared Optical Activity", ***Chemistry of Materials,*** 2020, 32, 476–488

(9) J. -Y. Kim, J. Yeom, H. Calcaterra, G. Zhao, J. Munn, P. Zhang, N. A. Kotov, "Assembly of Gold Nanoparticles into Chiral Superstructures Driven by Circularly Polarized Light", ***J. Am. Chem. Soc.*** 2019, 141, 11739−11744

(10) W. Feng, J.-Y. Kim, X. Wang, H. A. Calcaterra, Z. Qu, L. Meshi, N. A. Kotov, "Assembly of mesoscale helices with near-unity enantiomeric excess and light-matter interactions for chiral semiconductors", ***Science Advances*** 2017, 3, e1601159**.**

(11) L. Ohnoutek, J.-Y. Kim, J. Lu, B. J. Olohan, D. M. Răsădean, G. D. Pantoș, N. A. Kotov, V. K. Valev, "Third Harmonic Mie Scattering from Semiconductor Nanohelices", ***Nature Photonics*** 2022, 16, 126-133

(12) W. Choi, K. Yano, M. Cha, F. Colombari, J. -Y. Kim, Y. Wang, S. H. Lee, J. Kruger, André Farias de Moura, Nicholas A Kotov, "Chiral Phonons in Microcrystals and Nanofibrils of Biomolecules" ***Nature Photonics*** *2022*, 16, 366-373

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization